MILITARY SPECIFICATION

CLOTH, BUNTING, ACRYLIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an acrylic cloth.

1.2 Classification. The cloth shall be of one type only, in the following colors (see 6.2).

White	- 3032
Black	- 3227
Blue	- 3341
Green	- 3429
Yellow	- 3510
Red	- 3803

2. APPLICABLE DOCUMENTS

2.1 Government documents

*2.1.1 Specifications, standards, and handbooks Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sample Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.1.2 Other Government documents, drawings, and publications The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

Rules and Regulations
under the
Textile Fiber Product Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, D.C. 20580.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

Technical Manual of American
Association of Textile Chemists and Colorists

(Application for copies of the AATCC Manual should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, North Carolina 27709.)

3. REQUIREMENTS

3.1 Standard Sample. The finished cloth shall be equal to the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, the contractor shall furnish a sample of cloth for first article inspection and approval (see 4.3 and 6.2).

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3.3 Material (see 6.6)

3.3.1 Fiber. The fiber shall be first quality, 3 denier, semi-dull acrylic staple with a minimum staple length of 2 inches.

3.3.2 Yarns. The acrylic fiber shall be evenly spun and twisted into a two-ply yarn for both the warp and filling.

3.4 Color. The color of the finished cloth shall be as specified and shall match the standard samples.

3.4.1 Matching. The color of the finished cloth shall match the standard sample under artificial daylight having a color temperature of 7000 (+ 500) Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850 (+ 100) Kelvin.

3.4.2 Colorfastness. The dyed and finished cloth shall show fastness to crocking, laundering, weathering and sea water, and white cloth shall show fastness to weathering and sea water, equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is available, the dyed cloth shall show a minimum of "good" fastness to laundering, weathering, and sea water when tested as specified in 4.5. When no white standard sample is available, the finished cloth shall show good fastness to weathering and sea water. Crocking for all colors except white shall show a Munsell Value of not less than 8.5 when tested as specified in 4.5.

3.5 Physical requirements. The finished cloth shall conform to the requirements in Table I when tested as specified in 4.5.

Table I - Physical Requirements

Weight Oz./Sq. Yd. (g/m^2)	Yarns (per inch) (per Centimeter)		Breaking Strength (lb.) (Newtons)		Air Permeability ($ft^3/min/ft^2$) ($cm^3/sec/cm^2$)	Weave
Minimum	Minimum		Minimum		Minimum	
	Warp	Filling	Warp	Filling		
4.9 (166.1)	29 (11.4)	29 (11.4)	95 (422.75)	100 (445)	250 (127)	Plain

*3.5.1 Width. The minimum acceptable width of the finished cloth, inclusive of selvages, shall be as specified (see 6.2). The selvage shall be 1/4 (+ 1/8) inch (6.35 + 3.18 mm) wide. When shuttleless loom is utilized the width of the cloth shall be as specified (see 6.2) and shall be excluding the selvages.

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3.5.2 Shrinkage. The cloth shall be finished so as not to shrink more than 4.0 percent in the warp or 3.0 percent in the filling direction when tested as specified in 4.5. When a preshrinking process is used, it shall not be identified by name or trademark on the cloth, ticket or package.

3.6 Finish. The cloth shall be dyed, finished and heat set equal to or better than the standard sample.

3.7 Non-fibrous material. The total sizing, finishing and other non-fibrous material content shall not exceed 2.0 percent when tested as specified in 4.5.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous pieces, each not less than 40 yards (35.56 m) in length. Each length of cloth furnished shall be put up upon rolls, full width with the face side inward as specified in PPP-P-1133.

3.9 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.10 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.11 Face marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the work "FACE" at each end of the piece.

3.12 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, ($83.61 m^2$) when calculated as specified in Section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirments specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of Inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article of the finished cloth submitted in accordance with 3.2 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical and chemical properties in accordance with the methods specified in 4.4.

4.4 Quality conformance inspection. Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.5.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side. When the total yardage in the roll does not exceed 100 yards (91.40 m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.40 m), only 100 yards (91.40 m) shall be examined. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container, unless the number of shipping containers in the lot is less than the required number of rolls, in which case, all shipping containers shall be present in the sample. If less than 8 rolls in lot, all rolls shall be examined.

All defects as defined in Section III of FED-STD-4, which are clearly noticeable at normal inspection distances (3 feet) (0.91 m) shall be scored and assigned demerit points as listed in 4.4.2.2, except that only those slub and knots which exceed the maximun limits shown on Sears Fabric Defect Scales (see 6.4) "E" for slubs and "D" for knots shall be scored. No linear yard (0.91 m)

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(Increments of 1 yard (0.91 m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The lot shall be unacceptable if the points per 100 square yards (83.61 m^2) of the total yardage examined exceeds 35 points. The lot shall be unacceptable if the points per 100 square yards (83.61 m^2) of two or more individual rolls exceeds 45.0 points. If one roll exceeds 45.0 points per 100 square yards (83.61 m^2), a second sample equal in size to the first sample shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45.0 points per 100 square yards (83.61 m^2). Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 square yards. (83.61 } m^2\text{)}$$

4.4.2.2 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches (7.62 cm) in any direction	- one point
For defects exceeding 3 inches (7.62 cm), but not exceeding 6 inches (15.24 cm) in any direction	- two points
For defects exceeding 6 inches (15.24 cm), but not exceeding 9 inches (22.86 cm) in any direction	- three points
For defects exceeding 9 inches (22.86 cm) in any direction	- four points

The following defects, when present, shall be scored four points for each yard (0.91 m) in which they occur:

Baggy, ridgy or wavy cloth
Overall uncleanness
Degree and character of finish not equal to standard sample
Poor dye penetration, mottled, streaky or cloudy
Width of cloth less than specified or width of selvage beyond specified tolerances
Bowed or bias filling 2 1/2 inches (6.4cm) or more when measured from a line 90 degrees from selvage edge to maximum deflections.

4.4.2.3 Examination for length.

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4.4.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards (1.82 m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of the rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

* 4.4.2.4 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any roll in the sample exhibiting uneven shade, shade variation side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.4.2.5 Examination for face identification, identification of pre-shrinkage process and noncompliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample have face identification missing from either or both ends, two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing, and marking complies with the Section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191 and AATCC Technical Manual wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for test purposes shall be 3-1/2 continuous yards (3.19 m) full width of the finished cloth. The lot size shall be expressed in units of one yard (.91 m). The lot shall be unacceptable if one or more of the sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table II - Test Methods

Characteristics	Requirement Paragraph	Test Method
Acrylic Fiber:		
Identification of fiber	3.3.1	see 4.5.1
Luster of fiber	3.3.1	Comparison w/standard 1/
Quality, denier, staple length	3.3.1	1/
Yarn Ply	3.3.2	Visual 2/
Weight	Table I	5041
Yarns per inch	Table I	5050
Breaking strength	Table I	5100
Air permeability	Table I	5450
Weave	Table I	Visual 2/
Shrinkage	3.5.2	5556 3/
Non-fibrous	3.7	2611
Colorfastness to:		
Crocking	3.4.2	5651
Laundering	3.4.2	5610 4/
Weathering	3.4.2	5671
Sea water	3.4.2	AATCC 106-1972 5/
pH	3.10	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

3/ The cotton laundering procedure shall be used, except that the drying temperature shall be 150° to 170°F (66° to 77°C) and the drying time shall be 30 to 45 minutes.

4/ Only the stains on the acrylic fiber shall be evaluated.

5/ Test results shall be reported in accordance with Test Method 5610 of FED-STD-191.

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4.5.1 Acrylic fiber identification. Cut 0.5 gms of the sample unit into approximately 1/4 inch squares. Immerse in 50 mls of a 70 percent solution of potassium thiocyanate and raise to a temperature of 100°C; boil for 10 minutes. If the sample dissolves completely, the fiber shall be identified as acrylic. One determination shall be made from each sample unit and the results reported as "Pass" or "Fail".

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A or C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B or C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1133.

6. NOTES

6.1 Intended use. The cloth covered by this specification is primarily intended for use in the manufacture of signal flags.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When first article is required (see 3.2), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity and testing and approval of the first article.
c. Color required (see 1.2).
d. Width required (see 3.5.1).
e. Minimum acceptable length if other than specified (see 3.8).
f. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample address the procuring office issuing the invitation for bids (see 3.1).

*6.4 Fabric defect scales Fabric Defect Replica Kits are available from Sears Roebuck and Company, Department 817, (ATTN: BSC 23-29), Sears Tower, Chicago, IL 60684.

*6.5 Recycled material It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

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6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - NU

Preparing activity:
Navy - NU

Review activities:
Army - IH, MO
DLA - CT

Project No. 8305-0835

User activities:
Navy - OS,CG, SH

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-22775D	Cloth, Bunting, Acrylic

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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